ITEM-BASED COLLABORATIVE FILTERING

Goal: Predict the missing rating of **User A** for **Item-4**.

Dataset

User	Item-1	Item-2	Item-3	Item-4
User A	4	3	5	?
User B	5	5	4	3
User C	4	3	5	3
User D	2	4	2	3

Core logic: In item-based collaborative filtering, we do not first search for users similar to User A. Instead, we search for **items similar to Item-4**. Then we use User A's ratings on those similar items to predict User A's rating for Item-4.

Step 1: Decide what to compare

Since User A has not rated Item-4, we ignore User A while calculating similarity with Item-4.

So, to compare Item-4 with other items, we use only users who rated Item-4: **User B, User C, and User D**.

Item Pair	Use these users	Vectors used for cosine similarity
Item-4 vs Item-1	B,C,D	I4=[3,3,3]; I1=[5,4,2]
Item-4 vs Item-2	B,C,D	I4=[3,3,3]; I2=[5,3,4]
Item-4 vs Item-3	B,C,D	I4=[3,3,3]; I3=[4,5,2]

Simple classroom line: Item-based CF says: "Because User A liked similar items, we can estimate how much User A may like Item-4."

STEP 2: Calculate Item Similarities using Cosine Similarity

Formula

$$sim(Item\ X, Item\ Y) = \Sigma(Rating_X \times Rating_Y) / sqrt(\Sigma\ Rating_X^2 \times \Sigma\ Rating_Y^2)$$

Here, Item-4 is the target item. So we calculate:

1. Similarity(Item-4, Item-1)
2. Similarity(Item-4, Item-2)
3. Similarity(Item-4, Item-3)

A) Similarity between Item-4 and Item-1

Item-4 = [3, 3, 3] Item-1 = [5, 4, 2]

Numerator = (3×5) + (3×4) + (3×2) = 15 + 12 + 6 = 33

Denominator = $sqrt(3^2+3^2+3^2) \times sqrt(5^2+4^2+2^2) = sqrt(27) \times sqrt(45)$

Similarity = 33 / (5.196 × 6.708) = 33 / 34.856 = **0.947**

B) Similarity between Item-4 and Item-2

Item-4 = [3, 3, 3] Item-2 = [5, 3, 4]

Numerator = (3×5) + (3×3) + (3×4) = 15 + 9 + 12 = 36

Denominator = $sqrt(3^2+3^2+3^2) \times sqrt(5^2+3^2+4^2) = sqrt(27) \times sqrt(50)$

Similarity = 36 / (5.196 × 7.071) = 36 / 36.742 = **0.980**

C) Similarity between Item-4 and Item-3

Item-4 = [3, 3, 3] Item-3 = [4, 5, 2]

Numerator = (3×4) + (3×5) + (3×2) = 12 + 15 + 6 = 33

Denominator = $sqrt(3^2+3^2+3^2) \times sqrt(4^2+5^2+2^2) = sqrt(27) \times sqrt(45)$

Similarity = 33 / (5.196 × 6.708) = 33 / 34.856 = **0.947**

STEP 3: Select Similar Items and Predict Rating

Similarity Pair	Similarity Score
Item-4 with Item-1	0.947
Item-4 with Item-2	0.980
Item-4 with Item-3	0.947

All three items are strongly similar to Item-4. To avoid a tie issue between Item-1 and Item-3, we will use all 3 items for prediction.

User A ratings on these similar items

Item	User A Rating	Similarity with Item-4
Item-1	4	0.947
Item-2	3	0.980
Item-3	5	0.947

Prediction formula

Predicted Rating = Σ(Similarity × User Rating) / Σ Similarity

Prediction = [(0.947×4) + (0.980×3) + (0.947×5)] / [0.947 + 0.980 + 0.947]

Prediction = [3.788 + 2.940 + 4.735] / 2.874

Prediction = 11.463 / 2.874 = **3.99** ≈ **4.0**

Final conclusion

The predicted rating for **User A on Item-4** is approximately **4.0**. So, the system can recommend Item-4 to User A.

One-line difference from user-based CF:
User-based CF finds similar users. Item-based CF finds similar items.